

The Art of Solving Thermodynamics Problems

Chunk 2 — How to Plan Your Attack

Voiceover transcript (Aru-drafted)

2026-05-31

Slide 1. How to plan your attack

In Chunk 1 we argued that planning is most of the work. So let us look at how to plan the attack on a thermodynamics problem, step by step.

Slide 2. The five buckets

Every closed-system calculation for an ideal gas is built from five kinds of equation. Definitions, like enthalpy and the specific heats. Equations of state, which for an ideal gas are $Pv = RT$ and $u = c_v T$. Conservation of energy. The work and heat relations. And the process specifications. Keep these five buckets in mind: everything we write comes from one of them.

Slide 3. Thermo problems are puzzles

Solving a problem is then a puzzle. You know the rules — the equations in the buckets. You identify what you are given, and what is unknown. Then you use the rules to deduce the unknowns from the givens, one step at a time. The game is to keep assembling equations until you have as many as you have unknowns.

Slide 4. The starting point — what fixes a state

Before we solve anything, one rule sets up the whole game. To find the properties at a state, you must first know two independent intensive properties there. This is the State Postulate. For an ideal gas it takes a particularly simple form: know any two of pressure, specific volume and temperature, and the third follows at once from $Pv = RT$. And because internal energy and enthalpy depend on temperature alone, with $u = c_v T$ and $h = c_p T$, once you have the temperature those follow too. So any two of P , v and T at a state unlock everything else there.

Slide 5. The whole game — state to state

That turns the whole problem into a chain. At each state you spot the two of P , v and T that you already know, and the equation of state gives you the third. The process joining that state to the next hands you one property at the next state, so you can fix it the same way. You move along, state by state, until P , v and T are known everywhere. Only then do you work out the energy changes, the work and the heat. Everything that follows is just that idea, carried out with care.

Slide 6. Our example

Let us plan and solve one problem together, the whole way through.

Slide 7. A typical problem statement

Here is the problem. Air in a pneumatic cylinder starts at 100 kPa and 300 K. It is compressed isothermally to a specific volume of $0.2 \text{ m}^3/\text{kg}$, then cooled at constant volume until 250 K. We want the net work and the net heat. Read like that, it is dense. Where do we even start?

Slide 8. Step 0 — which ideal gas?

Step zero, every single time: identify the working fluid, and which ideal gas it is. Then write down its constants from Table A-2: the gas constant R , the specific heat c_v , the constant-pressure heat $c_p = c_v + R$, and their ratio k .

Slide 9. Step 0 for our example

For our problem the fluid is air. From the table, $R = 0.287$, $c_v = 0.718$ and $c_p = 1.005$, all in kJ per kg per kelvin, and $k = 1.4$. Notice too that the words “pneumatic cylinder” are telling us this is a closed system.

Slide 10. Step 1 — states and processes

Step one: find the key states, number them, and count them. Then identify the canonical process joining each pair of states. Every process we meet is one of four kinds: constant volume, constant pressure, constant temperature, or isentropic — a quasi-static adiabatic.

Slide 11. Step 1 for our example

Reading our problem this way, the state information is in blue and the process information in red. There are three states. State 1 is at 100 kPa and 300 K. One to two is isothermal. State 2 is at the specific volume 0.2. Two to three is constant volume. And state 3 is at 250 K.

Slide 12. Step 2 — the arrow diagram

Step two: lay the states out left to right as an arrow diagram. Above each state write the pressure, volume and temperature you know, and a question mark for the ones you do not. Under each arrow write the process and its $P(v)$, together with any heat or work you are told.

Slide 13. Step 2 for our example

Here it is for our problem. State 1 has its pressure and temperature; its specific volume is unknown. The process to state 2 is isothermal, and we are given the specific volume there. The process to state 3 is constant volume, and we are given its temperature. Everything else is a question mark, including the energy quantities under each arrow.

Slide 14. Step 3 — the P - v sketch

Step three: sketch the process path on a pressure-volume diagram. The arrow diagram fixed the topology, the order of states; the P - v sketch shows the shape of each path. It is your check that the processes join up sensibly. And remember the work for each process is the area under its curve.

Slide 15. Step 3 for our example

For us, the isothermal compression is a curve down to a smaller volume, and the constant-volume cooling is a straight drop at fixed volume. The areas under those curves are the boundary works.

Slide 16. Stage 1 — fix every state

Now the calculation has two stages. Stage one: fix the pressure, volume and temperature of every state. Start at the state where you already know two of the three; the ideal-gas equation of state gives the third. At each next state, the given information supplies one property, the process supplies a second, and the equation of state gives the third. Do not put in numbers yet; just write the equation under each state.

Slide 17. Stage 1 for our example

Here is that reasoning for our problem. At state 1 the pressure and temperature are given, and the equation of state gives the specific volume. At state 2 the volume is given, the isothermal process tells us the temperature equals state 1's, and the equation of state gives the pressure. State 3 follows the same way through the constant-volume process. The blue lines come from the equation of state, the red from the process.

Slide 18. Stage 2 — work and heat

Stage two: with pressure, volume and temperature now known, the energy change, work and heat follow. For every process, three equations: the change in internal energy is c_v

times the temperature change; the boundary work is the integral of $P dv$ along that path; and the heat is the internal-energy change plus the work.

Slide 19. Stage 2 — work for each process

The only equation that differs between processes is the boundary work, and you have seen these derived in the lectures. For constant volume the work is zero. For constant pressure it is P times the volume change. For isothermal it is RT_1 times the natural log of the volume ratio. And for isentropic it is c_v times the temperature drop. Write the right one under each arrow.

Slide 20. Stage 2 for our example

Putting those under our two arrows: for the isothermal step the internal energy does not change, the work is $RT_1 \ln(v_2/v_1)$, and the heat equals that work. For the constant-volume step the work is zero, and the heat is just the internal-energy change.

Slide 21. Step 5 — finish

Finally, add the legs together, keeping their signs. The net work is the sum of the boundary works; the net heat is the sum of the heats. From there you can get anything else the problem asks: an efficiency, a required mass, and so on.

Slide 22. Step 5 for our example

For our problem the net work is the work of the isothermal step plus the work of the constant-volume step, and the net heat is the sum of the two heats. This single diagram now holds the entire calculation plan.

Slide 23. The plan in one line

So the whole plan, in one line: identify the ideal gas; find the states and the processes; draw the arrow diagram; sketch the P - v diagram; fix pressure, volume and temperature; then the internal-energy change, work and heat; and finish.

Slide 24. You can now attack any problem

And that is it. The same five buckets and the same steps handle any perfect-ideal-gas problem, whether it is an engine, a pneumatic system, or a full cycle. The planning did the hard part; the arithmetic that follows is just bookkeeping.